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CERA: a collaborative environment reference architecture for interoperable CWE systems

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Abstract In this paper, we present the collaborative environment reference architecture (CERA) with the aim of supporting collaborative work environment (CWE) interoperability. The vision of CERA is to support users who are engaged in common collaborative spaces with similar work processes to work and collaborate seamlessly together, despite their use of proprietary CWE tools and systems. The underlying CERA concepts, design principles, and models are discussed, as well as the architectural decisions made as a result of the extended requirements analysis exercise. Furthermore, we present results from the Ecospace (http://www.ip-ecospace.org/) project as an example of a CERA instantiation which focuses on facilitating users collaborating across different CWE systems, namely BSCW, NetWeaver, and BC. We conclude with future research and implementation directions.

Keywords E-collaboration · Service oriented architecture · Semantic web · Interoperability · Reference architectures · Collaborative work environments

1 Introduction and background

In the area of collaborative work environments (CWEs), there are numerous software platforms that are presented as the optimal solution to address the needs of users. Such platforms support a wide variety of collaborative functionalities such as document management, project workspaces, shared TODO lists, and calendar management. CWE platforms providing these functionalities, such as Microsoft Sharepoint, basic support for corporative work (BSCW), business collaborator (BC), and SAP NetWeaver, are already successfully exploited by many organisations. However in current business activity, collaboration and cooperation not only take place within a single organisation, but also increasingly between organisations; a situation that raises some fundamental challenges:

- The cooperating organisations may use different CWE platforms that are non-interoperable with one another, even though there may be significant functional overlap amongst them;
- Organisational access policies or licensing restrictions make it infeasible to open the local CWE platform to external users beyond the organisational boundaries;
- Even in cases where an organisation might open its internal CWE platform to external users, there will be a significant cognitive load on those users to familiarise themselves with a new environment with alternative terminology and GUI organisation. This cognitive load increases for users who are involved in many cross-organisational cooperation processes supported by different CWE platforms.

The usual architectural approach in CWE platform development has been to encompass the broader possible area of the entire CWE problem domain, resulting in solutions which impose on users a monolithic system that attempts to fulfil all CWE requirements. This approach is based on the premise that users prefer to have a single platform to collaborate with

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their counter-parts across multiple work contexts, as opposed to dealing with multiple CWEs. This enables them to focus on the nature of the collaborative tasks and their decomposition into finer granular steps. Albeit true that users prefer a single platform, a monolithic approach does not take into account that each user may already work with a distinct CWE platform. Additionally users have varying functionality needs and preferences regarding CWE platforms, which a single solution cannot satisfy. This diversity, together with the lack of interoperability across the different CWE solutions, raises strong barriers for collaboration, which leads users to:

- Adopt the simplest, common communication–methods, such as e-mail and file transfer, due to the fact that these technologies are based on standard and widely acceptable protocols such as POP, IMAP, SMTP and FTP, and
- Attempt to construct a custom solution from distinct and, most likely, incompatible building blocks of collaborative functionality, resulting in an overlap of functionality and severe data portability issues.

In both cases, the potential of CWE platforms for efficient cooperation is not utilised. Towards the goal of overcoming these problems, this paper presents the collaborative environment reference architecture (CERA). CERA is a reference architecture with the aim of supporting interoperability between CWE systems. CERA deliberately remains technology-independent, meaning that different technical approaches may be adopted in its implementation. As an example of how CERA can be used as a real-world solution, we also present the Ecospace architecture, which is a specific implementation of CERA.

The CERA has been developed in the context of the IST Ecospace Integrated Project [1], which aims to develop CWE systems and tools capable of supporting the innovative work paradigms and processes of eProfessionals. Within this project, an eProfessional is defined as a professional whose business and tasks can only be achieved using modern cooperation technologies. These technologies enable an eProfessional to be part of a community and to be involved in distributed cooperation processes that were not previously possible. Being an eProfessional is not a profession of its own, but it exists in combination with a business profession such as a consultant, engineer, journalist, scientist, etc. An eProfessional is linked to a normal organisation through employment, but may also act in a self-employed way. To work remotely and/or in mobile environments is the norm for an eProfessional and he/she is often involved in many different projects within communities, projects, and with external partners in different organisations. Often these projects are constructed around highly complex and creative tasks that require a high coordination effort. Unforeseen issues may arise, requiring access to previously unknown information/knowledge. Thus tasks and processes of varying length and complexity, involving a variety of support tools, may be difficult to plan in advance. They require dynamic, ad hoc collaboration with different people and groups. eProfessionals are both a result of flexible, new business models and also the necessary pre-requisite for their implementation. Applying the 'form follows function' philosophy,[1] a collaboration environment that can adequately support the needs of an eProfessional must provide services on demand, based on the flexible work tasks of the user, i.e. the tools follow the cooperation needs of the eProfessional in (virtual) communities.

The vision of a networked, user-centric workspace [2] permits the emergence of dynamic CWEs that adapt to context and enable people not only to work from anywhere and anytime, but also to boost their collective creativity and to facilitate collaborative knowledge-generation. CERA's underlying design principle is to have an open, layered architecture, where only the foundational layer is mandatory for achieving a minimum level of interoperability and all others are optional for additional functionalities and coverage. This core architectural principle addresses the traditional pitfall of monolithic solutions, delegating to the system developers the decision of productivity versus flexibility by adopting the appropriate layers. CERA facilitates the creation of virtual CWEs where different CWE systems and tools can interoperate, enabling users to have their own personalised configuration of solutions that combines legacy functionalities. For developing the Ecospace architecture, CERA was used as the blueprint, while the most appropriate technologies that met the Ecospace requirements were adopted. As such, the Ecospace architecture is one of the many possible technical instantiations of CERA.

The paper is organised as follows: in the following section, we present related work concerning proposed referenced architectures for CWEs, focusing on the limitations of current commercial approaches. Section 3 outlines a motivating scenario and the results of the requirements elicitation phase that was run within the project. In Sect. 4, the main design principles that drove us while drafting the reference architecture are presented. In Sect. 5, CERA is described in detail. In Sect. 6, two tools that have been developed following the CERA architectural guidelines are presented. Section 7 outlines the major findings of our evaluation on CERA and, finally, in Sect. 8, we conclude and provide some future direction of work.

[1] Louis Sullivan: http://en.wikipedia.org/wiki/Form_follows_function.

2 Related work

The CWE reference architecture, CERA, facilitates seamless, dynamic and creative collaboration across teams, organisations and communities. In this section, we review the most important related projects and how their work compares to the approach presented in this paper.

Attempts to address the heterogeneity issues in computer supported cooperative work (CSCW) environments can be traced back as far as 1993 [3]. The MOCCA project aimed to develop common concepts for open collaboration systems [4]. The MOCCA approach identified a set of models, which collectively specified the collaborative environment. These were an information model, an organisation model, a workspace model and a rooms model. To a certain extent, these models may be mapped to the CERA content, user, context and process model. However, the CERA approach builds on these basic models by including further layers, e.g. the service layer.

Today, there are many research projects and commercial products active within this area [5]. Three of industry's biggest players, IBM, Microsoft, and SAP, provide mainstream collaboration support to eProfessionals. IBM offers the IBM Workplace, which combines many in-house collaborative applications, e.g. Lotus Notes/Domino. Microsoft offers Microsoft Windows SharePoint to support project management and eProfessional collaboration. The SAP NetWeaver platform allows users to design, build, implement, and execute new business strategies and processes by recomposing existing systems and combining them with modules that come with the product. Web-based collaboration solutions are also available. They include BSCW, BC, and the Polymedia Content Management Suite; all of which provide web-based shared workspaces for project management in collaborative environments. However, limitations of current commercial tools with respect to the support of eProfessional collaboration are as follows:

- *Lack of interoperability between tools of different eProfessionals*: Tools are designed based on the assumption that all people use the same toolset. In many settings of knowledge worker collaboration, especially in the case where these knowledge workers work for different organisations, this is not a valid assumption.
- *Lack of interoperability between tools of one knowledge worker*: Due to this lack of interoperability, it is difficult, or even impossible, for eProfessionals to integrate functionality and have one conceptual collaborative working environment. As a result, eProfessionals often have to perform a series of small actions on different tools, for instance to initiate a video conference with a group of peers to discuss a document. If the various tools would be integrated in a smart way, such a process could be supported by a single action in an integrated environment.
- *Lack of focus on teams of professionals and their strategies in collaborative work to achieve common objectives*: For such strategies to be successful, it is important to have tool support that helps teams of professionals to gain insight in the team strategies and individual strategies, provides relevant status information (for instance on the status of activities, task dependencies, bottlenecks, and awareness about the actors involved), and helps teams to coordinate their activities in complex team settings.
- *Lack of support to reduce information overload, for instance by providing clues about the relevance of material*: eProfessionals feel insufficiently supported by current tools in their efforts to reduce information overload and to provide the right context information with shared material (for instance, "Why is this material relevant for me?").
- *Lack of support for collaboration processes as they happen*: many of the existing tools seem to be designed based on the assumption of structured, predictable collaboration processes. However, knowledge worker processes are dynamic, ill-structured and creative; any tools that are designed based on assumptions regarding the structure of a collaboration process may hinder this process. eProfessional collaboration tools should therefore be able to support collaboration processes as they happen.

These problems continue to persist with regards to the design of CWEs as long as a technology driven approach is taken. An alternative approach is to assess and study the behavioural patterns of collaboration between e-professionals and organisations, thus determining the actual needs and requirements to be supported by CWEs.

In addition to industry's attempts to provide a solution for eProfessionals collaborating, there are currently several research projects working in this area, such as CoSpaces, C@R, DiFac, POPEYE, and inContext. CoSpaces aims to develop organisational models and distributed technologies supporting innovative collaborative workspaces within the field of design and engineering [6]. The CoSpaces Collaboration Framework provides workspace applications for the specific domain of product design, maintenance, and constructibility processes, whereas CERA targets the CWEs of eProfessionals. Similar to CoSpaces, both DiFac [7] and POPEYE [8] are collaborative projects addressing specific domains. Comparable to the Ecospace project is C@R, as it considers a service based approach in its architecture design [9]. The inContext approach is most compatible to CERA's approach, because of its definition of a set of basic services as well as the consideration of the composite services [10]. Although C@R also considers a service based approach it differs in that it defines a bus system that serves as a middleware between the services and the client. CERA does not require such an intermediate bus and thus enables a more flexible and ad hoc interaction between different services. This has been proven by the interaction of shared workspace systems such as BSCW and BC using basic CERA services, supporting connectivity in addition to the content model, which enables information exchange.

A non-traditional research area for CWE, but highly relevant, is that concerning social semantic desktops (SSD). The SSD paradigm applies the ideas of the Semantic Web, to the problem of desktop-application interoperation. NEPOMUK aims to develop a solution for extending the personal desktop into a collaboration environment which supports both the personal information management and the sharing and exchange across social and organisational relations [11]. It also provides a standard reference-architecture for developing SSD systems. Although both NEPOMUK and CERA address resource sharing between users and task management, CERA specifically aims to provide a semantic, personalised, virtual CWE that facilitates collaboration between teams of eProfessionals whereas NEPOMUK focuses more on extending the personal computer into a collaborative environment.

3 CWE scenario and requirements

3.1 User scenario

In this section we present a simple motivating-scenario for the development of a CWE reference architecture, illustrating the benefits of addressing the current CWE interoperability issues. In this user scenario, there are three main actors: Alice, Bob, and Jane.

> Alice's IT firm has won a contract to develop a new payment-system for the multi-national, medical-device company HospiCare. She hires two consultants: Bob, an accountant; and Jane, a human resources (HR) specialist. Essential to the success of the project is the collaboration and coordination of all three professionals; however each typically uses a different CWE-platform for collaboration purposes. Thanks to CERA, interoperability between these distinct platforms is facilitated. During the first week of the project, Bob is based at the head-office of HospiCare in London. He is analysing the current payment scheme and identifying areas to be improved on. In order to record his findings, he creates a 'HospiCare Contract' workspace on his BSCW environment, which he updates regularly. At the same time, Jane is gathering requirements from management at HospiCare's main manufacturing plant in Dublin. She creates the workspace 'HospiCare' on her preferred CWE platform: BC. As well as recording the requirements in her own workspace, she wishes to incorporate Bob's findings into her interviews with management. Using the Workspace Synchronisation tool, Bob's work, stored in a BSCW workspace, may be imported by Jane into her BC workspace. Bob's remote storage is transparently shadowed as local storage in Jane's CWE system, creating a logically unified document repository.
> Alice wishes to touch-base with Bob and Jane and discuss some of the emerging requirements. As such, Alice schedules the meeting in her calendar, which implies accessing the calendars of Bob and Jane across the different CWE platforms, before identifying a common available slot. Once a time-slot has been found, Alice enters the meeting in her calendar and the event is automatically included in all the other calendars. The rest of the team is notified via e-mail and/or SMS message, depending on their preferences and current context. The collaborating eProfessionals are not based in the same location at the time of the meeting, so they use a video-conferencing tool, which offers audio/video functionality as well as desktop and CWE-content sharing.

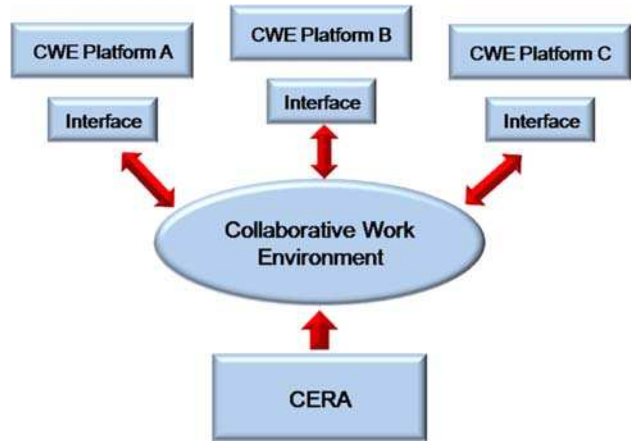

Fig. 1 Conceptual diagram representing the approach of CERA in ECOSPACE

The above scenario reflects how collaboration should take place amongst eProfessionals from multiple organisations, each with their own CWE platform.

In such settings, the strategy of adopting a single CWE platform is not feasible. The alternative is to create a logical CWE where each particular CWE platform is integrated. This was the approach taken in Ecospace, resulting in the conceptual diagram of Fig. 1. Ecospace proposes having a single logical CWE that each CWE platform integrates into by supporting the relevant interfaces for required collaborative functionality across the different solutions. These interfaces and their relationships are defined by CERA.

3.2 Requirements overview

Within the Ecospace project, we ran an extensive requirements elicitation and engineering phase based on the support of the three project living labs, each representative of different distinctive domains:

- Media Domain.
- Engineering Domain.
- Communities Domain.

Here we briefly present the aggregated lists of user and technical requirements. A typical use-case for each requirement is described in order to put the requirement in the right context. More details and documentation can be found in [12].

The user requirements gathered during the elicitation phase, shown in Table 1, drove the engineering of technical requirements from an architectural perspective. This design exercise resulted in the following list of technical requirements illustrated in Table 2.

In addition to the above functional requirements, non-functional technical requirements have been identified, such as scalability, reliability, exception-handling, availability, etc.

4 CWE design principles

4.1 Collaboration meta-model for CWE

A growing body of research demonstrates that eProfessionals are not confined to a single work context and multitasking across multiple projects and various initiatives is common [13, 14]. In [15], the authors attempt to formalise the concept of multitasking across multiple work spheres, where each sphere corresponds to a particular work context with its own set of tasks and people collaborating together. We combine the concept of work spheres with the key conceptual elements of e-collaboration proposed by [16], namely:

- individuals involved
- mental schemas possessed by individual (knowledge)
- collaborative task
- physical and social environment

We use these fundamental concepts to present our top-level model for collaboration which consists of four elements, as shown in Fig. 3. These are:

- User (who)
- Content (what)
- Process (how and when)
- Context (where)

The Content and Process Element form together the Collaboration Space. We have included the concept of work sphere here by defining the personal collaborative space (PCS) as the intersection of a User's work sphere with a collaborative space. A User may refer to an individual user or an organisation. Also, a User may have multiple PCS as they may participate in multiple collaborative projects.

Adopting a people-centric approach [2], Fig. 2 reads: The User participates in a collaborative space. The Collaborative Space is the virtual space associated to a particular project or initiative where people come together to collaborate. A User may be involved in multiple PCS (i.e. projects). In there, the motivation/goal of the collaboration is determined.

Each collaborative space has:

- Content: the data and metadata that populate the collaborative space.
- Processes: collaborative functionality that take place in the space.

The User and the Collaborative Space live and evolve within a Context. This is the external physical and social environment that affects the User and/or the collaboration space.

4.2 Key design principles and technology paradigms

There are two key design principles that were foundational in the design of CERA:

- *Activity Oriented.* The aim is to focus on the functionality that is necessary to support collaborative activity within a group of e-professionals and to avoid application driven design, as it does not lend itself to reuse. It also results in a sub-optimal organisation of work and enforces a technology driven approach of what a collaborative task is and how it should be performed.
- *Open Layered Architecture.* This means that the reference architecture is based on layers, but their adoption is optional, with the exception of the foundational layer that establishes the common denominator necessary to promote interoperability between different CWE platforms.

In addition, although CERA remains technology independent, two software paradigms influenced the design:

- *Service Oriented Architecture (SOA).* All functionality to be shared amongst CWE platforms should be exposed with a well-defined interface as services.
- *Semantic Web.* All collaborative functionalities, content, actors, and context should be semantically annotated, thus facilitating discovery, composition and execution, as well as data interoperability.

Table 1 User requirements

Requirement	Description	Use-case
Open networks	Collaboration teams should be able to have a dynamic lifecycle with people leaving and joining throughout the duration of the associated activities.	Ann has recently joined project A; she is given full user access to its CWE platform. Ben has moved from Project A to Project B; he is removed from Project A's CWE platform and added to Project B's. Tom has been promoted to manager of Project C; his CWE platform role is changed accordingly, so that he now may access all data in the CWE platform
Integration of function-alities	Users should have a single application that provides all the required CWE functionality.	Ann wishes to organise a video-conference. This involves arranging a suitable date and selecting a suitable video-conferencing tool that facilitates video-chat, text-chat, a common whiteboard, and visibility of each other's desktop screens. Ann enters the desired functionality of her video-conferencing tool and all matching tools are returned to her
Single sign-on	Users should be able to sign-on once to access all the different CWEs.	Tom works on Project A, Project B, and Project C. He wishes to access documents from each project, but does not wish to perform the log-in process three separate times
Informationinte-gration	Users should be able to share and combine information irrespective of the CWE platform where it resides.	Tom wishes to integrate the member data from Project A and Project C, so that he may see who participates in both projects and what their role in each project is
Context-awareness	The CWE platform should support user and system context awareness.	Ann wishes to contact Ben privately and discuss a point he made on a Project A forum. The CWE platform identifies that they are both online and initiates an IM chat. If Ben was not online, the platform may have initiated an email
CWE personalisation	Users should be able to adapt and configure the work environment with little effort.	Ben has specified a couple of particular forum threads in which he is interested. After logging-in, he is presented with all new items on these threads
Semantic search	The CWE platform should provide advanced searches based on data semantics.	Ann performs a search for all users that have the role of administrator or manager. In order to retrieve accurate results, all user data must be semantically annotated with a 'has_role' attribute
Role management	It should be possible to efficiently manage groups of users and assign roles to users.	Ben has the role 'user' in Project B, meaning he has normal access rights. Ann has the role 'user/administrator' in Project B, meaning she has normal access rights, plus configuration rights. Tom has the role 'manager' in Project C, meaning he has full access rights, and may view, write, and delete items, as he sees fit
Activity orientation	The CWE platform should be oriented towards user activities, as opposed to application-oriented.	Tom wishes to establish a brainstorming session concerning Topic X. This involves contacting all members that are interested in Topic X, arranging a suitable time, setting up a meeting, and recording the results. The activity of 'establishing a brainstorming session' may involve one or more tools, which should be provided by the CWE platform
User-friendliness (non-functional requirement)	The interface should have a very low impact on the cognitive load of the users.	Ben is new to Project B and needs to get up-to-speed with the project details quickly. He therefore requires a platform interface that he can easily navigate and use efficiently
QoS (non-functional requirement)	The CWE platform is required to be available, robust and responsive, whilst supported by access control according to roles.	As a manager, Tom often needs to access CWE platform information reliably and without delay, e.g. if he is in a meeting, giving a presentation, on a mobile device, etc. Therefore a CWE platform should be available, robust, and responsive

5 Collaborative environment reference architecture

In this section, we present CERA as a CWE reference architecture. Moreover for each layer, we briefly present the specific Ecospace implementation with the models and technologies which have been adopted in the project as an example of CERA instantiation. CERA is a layered architecture and its overview is presented in Fig. 3. The overall architecture consists of six layers. They are:

1. *Semantic Infrastructure Layer*: Represents models, metadata, and rules to be used by all the other layers in order to provide semantics to the entire solution.
2. *Basic Collaborative Service Layer*: Represents all available Basic Collaborative Services (BCS), which repre-

Table 2 System requirements

Requirement	Description	Use-case
Semantic discovery andsemantic composition ofservices	All collaborative functionalities should be available as services, semantically annotated and composable	System offers service discovery based on function, pre-condition and post-condition
Contextualisationmechanisms	The system should be able to adapt its functionalities to the context of the user.	System offers its functionality on mobile device or PC, depending if user is on the move
Personalisationmechanisms	The system needs to allow the user to tailor their personal workspace according to their needs	System provides event notifications as they happen, or as a daily digest, depending on user preferences
Recommendationmechanisms	The system provides guidance to the users whilst it is involved in their activities	System recommends that a user join a forum, if many of their colleagues are participating in that forum
Integration mechanisms	The system should facilitate the integration of independent, legacy CWE platforms, by supporting technical, semantic and process interoperability	A legacy CWE may interoperate with another, provided they both adopt CERA
Customisable and integrated GUI	The system should support an integrated, yet decoupled from the CWE platforms, interface as a common access portal	The system provides a much of its functionality in the form of widgets, which may be arranged in a widget portal according to user preferences

sent collaborative tasks that cannon be divided into smaller parts.

3. *Orchestration Layer*: Represents all available Composite Collaborative Services (CoCoS), which are sets of BCSs, executed in a defined order to provide value-added collaborative functionality.
4. *Collaborative Technologies and Tools Layer*: Represents the tools and collaborative technologies that facilitate collaboration.
5. *Interface Layer*: Represents the user-interface or front-end that is used by eProfessionals to interact with one or more CWE applications.

The Basic Collaborative Service Layer is the foundational layer that needs to be adopted in order to benefit from CERA, as this guarantees a minimum level of interoperability. The remaining layers are optional, but the more layers that are adopted, the greater the benefit, although complexity is also increased and flexibility is reduced. The remainder of the section presents the CERA layers in more detail.

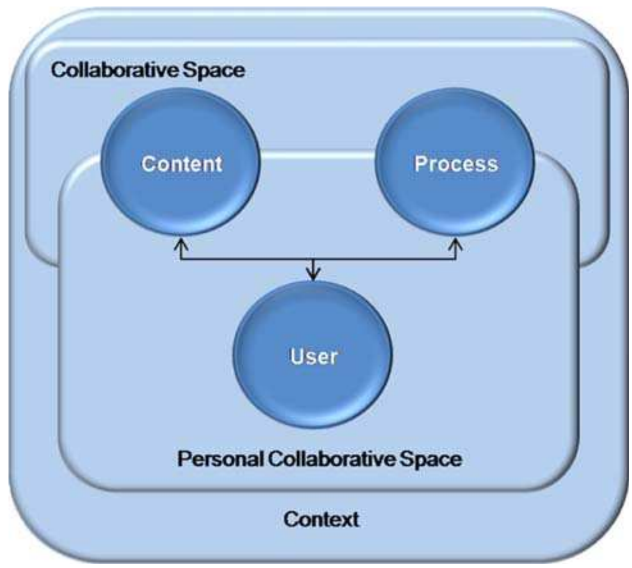

Fig. 2 Generic collaboration model

5.1 Semantic infrastructure layer

Based on the Generic Collaboration Model (Fig. 2), we have identified four core CWE models that provide formal descriptions, reference models and metadata for the CWE domain. These are:

- the *User Model,* which represents eProfessionals' characteristics,
- the *Process (or Service) Model,* which represents a CWE's functionality,
- the *Context Model,* which represents features of the dynamic eProfessional environment (e.g. temporal, spatial), and
- the *Content Model*, which represents metadata about the information stored and processed in a collaborative space.

The combination of the four models forms the CERA semantic infrastructure that permeates all the other components of the reference architecture. Each model enables certain type of functionalities. For example, the Context Model supports contextualisation mechanisms whereas the Content Model is used for annotating informational resources to facilitate discovery and data portability. Although CERA identifies the need for these four models to facilitate CWE interoperability, how these models are implemented, either conceptually or technically, are not specified. In this way, CERA remains

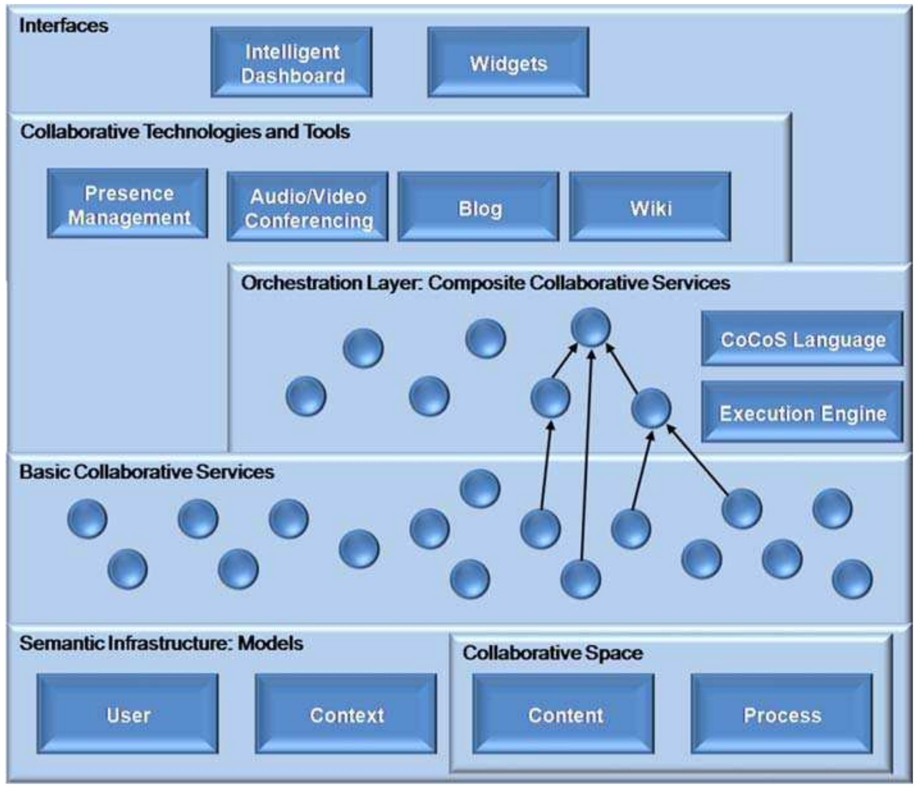

Fig. 3 The collaborative environment reference architecture (CERA)

technology independent and the decision of which approach to use is left to the CWE architects and developers.

However to illustrate the use of the CERA semantic models, we present how these conceptual reference models have been implemented in the Ecospace architecture. As the Ecospace architecture is an instantiation of CERA these models are examples of how the CERA models would look in a real implementation. The Ecospace models were selected based on the requirements of the project and the experience and knowledge of the project partners.

Because data portability and semantic interoperability amongst various existing CWE platforms have been important requirements for the Ecospace project, we focus here on the specification of the content model. Additionally, content plays an integral role in many of the tools developed as part of this project. Therefore, we briefly outline our approaches to the User, Context, and Process model in the Ecospace architecture, and describe in more detail the implementation of the Content model. The same model is used in Sect. 6 for presenting some prototypes and implementations.

5.1.1 The ecospace user model

In the Ecospace architecture, FOAF with Collaborative Vocabulary (CoVoC) [17] extensions is used as the basic eProfessional user model. CoVoC is a vocabulary developed within the project, which extends generic relationship–vocabularies, such as REL-X[2] and RELATIONSHIP.[3] It is used to identify the most prevalent types of acquaintances and social interactions in the collaboration of eProfessionals and, more generally, knowledge workers.

5.1.2 The ecospace process model

For the Ecospace architecture Process model, a semantic, web-service ontology is used, namely the SA-WSDL specification [18], for adding a semantic layer on top of the Web Service standards stack. In this way a uniform semantic description of CWE functionalities that are available as Web Services may be provided. More details can be found in Sect. 5.3.

5.1.3 The ecospace context model

The Context Model represents the characteristics that are related to the environment in which the user is placed and may affect the way a collaborative activity of any type is executed. An internal context model has been developed within the Ecospace project [19]. Additionally, we are currently

[2] http://www.dicom.uninsubria.it/~andrea.perego/vocs/relx.owl.

[3] http://vocab.org/relationship/.

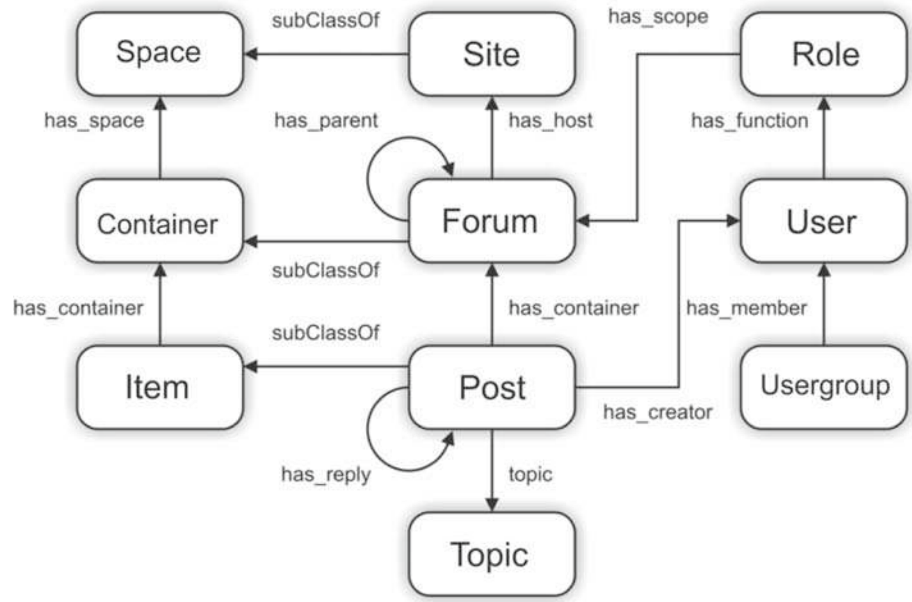

Fig. 4 SIOC-based CWE content model

examining preliminary versions of the Context Models developed elsewhere to try and determine whether and how they could be used in combination with and to enhance the Ecospace Context Model, e.g. the results from the InContext project.[4]

5.1.4 The ecospace content model

From a content perspective, all CWE applications and workspaces are semantic islands, as they use different vocabulary to describe similar information. However, it has been observed that, despite their differences, all these applications explicitly or implicitly model common elements. The objective of the Content Model is to represent these common elements, their underlying structure, and their semantics. The Content Model plays the role of a mediator or common language between different applications/tools. The goal and main functionality expected from the usage of this model is to semantically interlink CWE applications, thus enabling collaboration and information sharing across distributed and heterogeneous applications/tools.

In Ecospace, we use the semantically interlinked online community (SIOC) [20] ontology as a Content Model, arguing that Shared Workspaces can be perceived as a special type of online communities. SIOC is a framework that aims at connecting online community sites by using semantic web technologies and is currently a W3C submission. The core of SIOC is a vocabulary that contains concepts necessary to categorise information contained in online community sites. There are currently tools available for generating and using SIOC-annotated information. In Fig. 4 the main SIOC concepts and properties are presented.

Figure 5 shows the SIOC concepts and properties which have been identified in three CWE platforms, namely BSCW, BC and NetWeaver. A detailed description of how SIOC concepts are mapped to concepts and artefacts found in the three platforms is documented in the project wiki pages.[5] A brief description of the main concepts follows. A Community may consist of different CWEs joined by common topics, interests or goals. A Site is the CWE location, with Users creating Items in a set of Containers. A Container is an area in which content Items are stored. Each Container is hosted in one Site. There are different types of Containers, such as BookmarkFolder, projectDirectory, EventCalendar, AddressBook, Forum and Briefcase. An Item is any sharable object in a CWE. We can annotate an Item with user defined tags. A User is a person with an account on a CWE. By mapping the concepts of different CWEs platforms to SIOC, a unified semantic Content Model is created.

- Interesting CWE functionalities become possible with the adoption of a common Content model, e.g. SIOC in our case:
- Users are semantically connected with their multiple user accounts on different sites.
- Items from different CWEs that share the same topic can be semantically interlinked with each other.
- Distributed conversation and blogs are facilitated.
- A virtual container of all sharable objects can become available to a CWE user, irrespective of the CWE platform they use.

4 http://www.in-context.eu/.

5 http://www.cwe-projects.eu/ecospace/SIOCMapping.

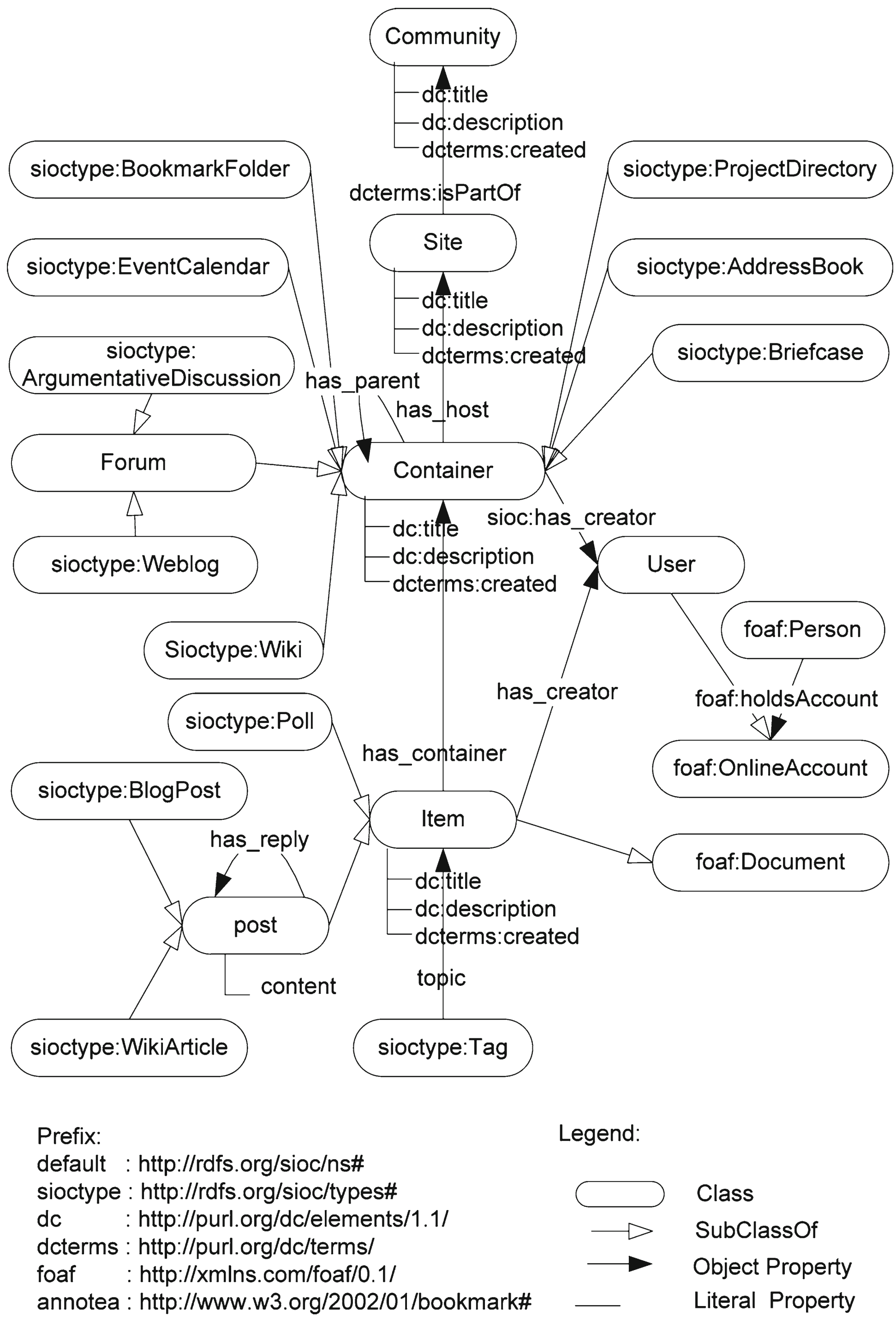

Fig. 5 SIOC concepts and properties for CWE

- Calendar events can be shared across different CWEs, so that users working on different CWEs are aware of all collaborative events that they are interested in.
- Relevant CWEs could be organised into unified and integrated collaboration communities.

Some concrete examples of the above benefits are demonstrated in Sect. 6, where tools based on using SIOC as the content model are presented.

5.2 Basic collaborative service layer

The Basic Service Layer consists of the set of basic collaborative services (BCS), common to all CWEs. It is the foundational layer that needs to be adopted in order to benefit from CERA, as its adoption guarantees a minimum level of interoperability. A BCS is defined as a service that supports a simple collaborative task that cannot be further divided

Table 3 List of BCS operations

No.	BCS	No.	BCS
1	Add	17	Invite
2	Accept	18	Uninvite
3	Get	19	Subscribe
4	Delete	20	Unsubscribe
5	Edit	21	Search
6	Decline	22	Copy
7	Start	23	Lock
8	End	24	Unlock
9	Recover	25	Export
10	Execute	26	Store
11	Send	27	Enter
12	Authenticate	28	Leave
13	Notify	29	Share
14	Check	30	Assign
15	Replace	31	Create
16	Reply		

into smaller tasks. This atomicity qualifies the BCS as the smallest building block of CERA. Another foundational BCS characteristic is that the BCS distil the base common functionality of any CWE, thus they are instantiated in different forms in each platform. The services that exist in this layer:

1. are implemented by some basic CWE technologies (e.g. e-mail, blogs)
2. become available to other application that may combine several basic CWE technologies (e.g. MS Outlook, BSCW video-conferencing)

Each BCS can support several operations of different types but with the same purpose of functionality. A complete list of the proposed BCS part of CERA is enumerated in Table 3.

These services have been documented using a template. The example of this template, for the 'Create' service is presented below. The aim of the Create BCS is to support the creation of any type of resource within CWE systems, irrespective of its complexity. This means that the Create BCS can be used for the creation of new blogs, new documents, users, or events. Within CWE systems there is always the need to create resources. However, instead of having a 'Create' per each type of resource and each particular nuance of a particular procedure call, careful refactoring was carried out to distil the basic functionality of the Create BCS. The concrete execution of the particular semantics of the Create BCS is determined at runtime according to the parameterisation of the service (what are the inputs provided).

ResourceID **Create** (platform, technology, operationType, set of pair-value)

Above is the signature of the Create BCS. The parameters of the service are as follows: platform indicates the CWE platform, e.g. BSCW; technology indicates the specific CWE technology, e.g. Blog; operationType indicates the operation within that technology, e.g. createBlog, createBlogLogin; and the set of pair-values indicate any other parameters necessary to the function.

Once a resource is created, the service returns the identity of the resource which can later be used to retrieve the resource from the system. The service handles the differences in creating resources between different systems.

Within the Ecospace project there was a decision to implement the BCS layer of CERA using Web Services technologies. Moreover, as we wanted to add a semantic layer on top of the service descriptions, we used the SA-WSDL specification. The main idea behind SOA is to wrap systems' functionalities and make them available as independent services with well-defined interfaces that can be called to perform their tasks in a standard way. Although SOA is not tied to a specific technology, currently the Web Services technology stack (i.e. SOAP, WSDL and UDDI) is commonly used to implement a SOA. The web service description language (WSDL) specifies an XML format that allows service interfaces to be described along with the details of their bindings to specific protocols. In Ecospace, we have chosen WSDL to describe BCS, but because WSDL follows the syntactic approach, we need an additional mechanism to extend WSDL with semantic information to automate or semi-automate certain tasks during or before service invocation. These tasks generally include service discovery, service composition, negotiation, filtering of discovered services, service selection based on some criteria, and service invocation.

Semantic Annotations for WSDL and XML Schema (SA-WSDL) defines how to add semantic annotations to various parts of a WSDL document such as input and output message structures, interfaces and operations. For example, SA-WSDL specification defines a way to annotate WSDL interfaces and operations with categorisation information which can be used to discover a Basic Collaborative Service (BCS) stored in the service registry. The annotations on XML-schema types can also be used to discover and compose BCS. The benefits of using SA-WSDL are that SA-WSDL is agnostic to a specific ontology or formalism to annotate the WSDL for the BCS. SA-WSDL provides the flexibility to extend the basic annotation framework according to the needs of various use-cases, e.g. modelling of pre-conditions and post-conditions for the BCS.

5.3 Orchestration layer: composite collaborative services

The Orchestration layer of CERA is built on top of, and is closely linked to, the BCS layer. In the Orchestration layer,

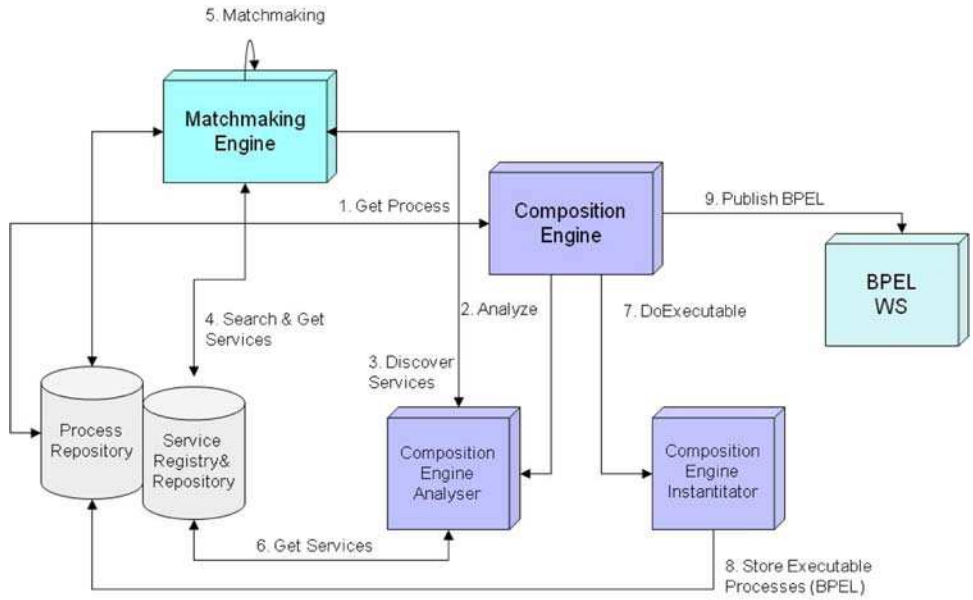

Fig. 6 The orchestration layer in the ecospace project

BCSs are orchestrated together to form Composite Collaborative Services (CoCoS) in a process usually referred to as Service Orchestration. A CoCoS can be defined as a composed set of BCSs that is executed in a predefined sequence to provide more complex, collaborative functionality than the BCS can offer. Compared to the BCS, in addition to the well-defined service interface (i.e. WSDL in the case of Web Service technologies), CoCoS are associated with a control flow that defines their execution logic and a dataflow that captures the data needed for the process. Moreover CoCoS interactions may span multiple applications and/or organisations, and result in a long-lived, transactional, multi-step process model.

A CoCoS can be specified at a high level by an eProfessional to model a complex collaboration process (e.g. setting up a meeting) as a set of composed functions implemented by a set of individual BCSs. The concrete implementation of the BCS's functional logic is the responsibility of the CWE developers. At an abstract level, a collaborative process template defines the publicly visible behaviour of some or all of the BCSs involved in the process. At a concrete level, all the necessary details need to be included to make a process template executable, e.g. business logic of the process, concrete data and service bindings, etc. Additional tasks (e.g. Service discovery) are required to create executable CoCoS from the CoCoS template.

From the implementation perspective within the project and based on the fact that in Ecospace, WSDL and SAWSDL are used to describe the BCS and CoCoS, we have chosen a WSDL-compliant orchestration model (i.e. WS-BPEL) for the CoCoS. The role of the WS-BPEL in Ecospace has been to enable an eProfessional to describe his collaborative process at both abstract and concrete levels. WS-BPEL is designed to fit naturally with the Web Services stack (i.e. implementation of the BCS layer).

Although, WS-BPEL allows describing abstract as well as concrete processes, an additional mechanism is required to generate an executable process out of an abstract process definition. This mechanism is usually known as Service Orchestration. Service Orchestration may be defined at design time (i.e. static orchestration) or at run time (i.e. dynamic orchestration). In the Ecospace project, a hybrid approach is taken to perform Service Orchestration, i.e. predefined abstract CoCoS templates are created at design time. The predefined CoCoS templates describe the overall process flow, but do not necessarily bind each process step to a specific basic service invocation. Therefore, we need some additional steps to create an executable CoCoS from an abstract CoCoS template. These steps are highlighted in Fig. 6.

The component responsible for Orchestration is known as Orchestration or Composition Engine. The first step in the orchestration process is to process the particular CoCoS template that can fulfil an eProfessional's collaboration task. Once it is processed, the Composition Engine sends it to an analyser which analyses the process template to identify which services are required to execute the process. A service discovery request is made to the matchmaking or discovery engine to discover all the required services to execute the process. The matchmaking or discovery engine implements the matchmaking algorithm based on the service's pre and post-conditions. Once the list of services is identified, it is then returned to the composition engine, which gives this information to a process-instantiater. The process-instantiater uses XSLT to transform the abstract process definition into an executable process definition using WS-BPEL.

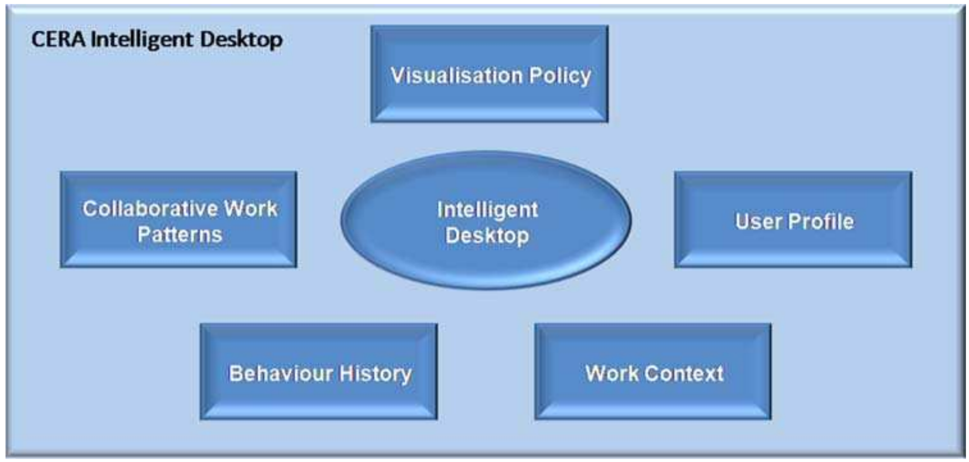

Fig. 7 Conceptual architecture of CERA intelligent desktop

Table 4 CWE technologies

No.	CWE technology	No.	CWE technology
1	Application sharing	18	Polls/surveys
2	Audio conference	19	Presence/availability
3	Blog	20	Resource management
4	Calendar	21	Rights management
5	Chat	22	Semantic annotation
6	Context management	23	Shared workspaces
7	Content management	24	Statistical
8	Control version	25	Synchronous multiuser editing
9	Database	26	Syndication
10	Distribution list	27	Task management
11	E-mail	28	Tracking/localisation
12	Forum	29	User management
13	Group management	30	Video conference
14	Instant messaging	31	Voting
15	News	32	Whiteboard
16	Notification/alert	33	Wiki
17	Phone	34	Workflow

5.4 Collaborative technologies and tools layer

At this layer, tools and technologies that facilitate collaboration can be found. Technologies refer to general application domains with a specific functionality, whereas tools refer to specific applications, which may exploit the functionality of one or more technologies. These tools both consume and make available basic and composite collaborative services, by invoking CoCoS and BCS in lower layers to provide the required functionality. Tools may be designed as either standalone applications or components in CWE platforms with broader coverage and scope.

Within the Ecospace project, and through analysing the various functionalities of existing CWE platforms, we identified thirty-four different CWE technologies, which are enumerated in Table 4. More information and documentation for these different tools can be found in the project wiki page.[6]

5.5 Interface layer

As pointed out by [21], information and technology innovation is becoming more complex and the challenge is achieving a clear design that unburdens the user's cognitive load making the most efficient use of their time. This is clearly evident with the plethora of tools and possible configurations of a CWE, thus a key user requirement of the interface layer is to provide a personalised environment for every e-professional based on their preferences, work patterns and their current shared work context.

In CERA, the intelligent desktop has been proposed as a CWE interface framework. It is imbued with semantic intelligence that enables the dynamic reconfiguration at run-time, thus adapting to both the work context as it changes and the work pattern of the user. The interface dynamicity may be automated or it may require the approval of the user, depending on the user's preferences. The conceptual architecture of the CERA intelligent desktop is captured in the schematic diagram of Fig. 7.

The desktop maintains the profile of the user and builds a history of their behaviours when using CWE. This information is complemented by the monitoring of the current work context that is assessed against the Collaborative Work Patterns defined by the appropriate ontology. The continuous analysis of the four data sources provides the means for the intelligent desktop to assess the best tools and services to support the work undertaken by the user. The result of the analysis is a priority list that the intelligent desktop uses as input for contextual cueing [22] that affects the visual management of the various tools/services based on focus, e.g.

[6] http://www.cwe-projects.eu/ecospace/CWE_Technologies.

what is the immediate task being done, and context, e.g. what is the current CWE, who is part of it, what is the user's role.

The visualisation based on focus and context takes a similar approach to the fisheye concept of website navigation, such as [23], where the degree of interest has an impact on the focal display of information/data. With the CERA intelligent desktop, the concept is applied to the tools/services to be used by CERA the current session of the CWE. The fundamental abstraction of the CERA intelligent desktop is the use of a window through which a user may interact to receive the output of the particular service/tool and provide the necessary input. As such, the area of the desktop can be divided into different areas of interest, which by default are two:

- *Primary*. This corresponds to the main area where the focus of the user resides and normally corresponds to the window/display of the main tool/service currently being used.
- *Secondary*. This corresponds to an area of the desktop that does not reside in the focus of the user, but contains the information concerning tools/service of potential relevance to the user.

The actual visual management of the areas depends on the visualisation policy associated to the CERA intelligent desktop, which may be configured by the user offline. The following two examples demonstrate different approaches to the visualisation policy:

- One of the simplest visualisation policies consists of attributing the Primary focal area to the window or display area of the tool/service that the user is currently interacting with. This is supported by a sidebar (Secondary focal area) that contains a tree structure of potential candidates for the primary focal area.
- A more sophisticated approach is to consider the use of two graphical layers, one for the Primary and another for the Secondary, with the former overlapping the latter. The Primary area of interest may correspond to either a single tool/service or a small set of tools/services of relatively similar importance to the user. The Secondary area of interest has large thumbnails of the different tools/services of potential relevance to the user. To distinguish between the levels of relevance, the contextual cues are used to modify the alpha channel of the thumbnails.

The granularity of areas of interest is configurable, so more areas may be considered, but these may conceptually be regarded as refinements of the Secondary area of interest. For example, one may consider an area of interest as one with the most recently used tools/services and another with the services/tools of potential relevance.

6 Prototypes and implementation

Within the Ecospace project a large number of collaborative tools have been developed [24]. Here we present two of these tools as an example of how CERA has been instantiated and what benefits became.

6.1 SIOC Xplore widget

The SIOC Xplore widget enables users to perform cross-platform browsing and querying in a unified way. It acts as a single-entry point to data, originating in distinct CWE platforms. Using the widget, queries such as 'find all items that Mary has created since 01-04-08' or 'find all users that have edited files pertaining to deliverable 4.5' will yield links to all the relevant informational artefacts, located on any of the specified CWEs, that meet the search criteria. This functionality alleviates the user from having to log-on and perform searches at each of the CWEs, and manually having to correlate the search results. The widget interface is shown in Fig. 8.

However, in order to facilitate data interoperability and enable the SIOC Xplore Widget to execute queries, correlate results, and present results to the end user, it is necessary to structure CWE data in a homogenous fashion based on the CERA Content Model. As discussed in Sect. 5.1.4, in Ecospace this model is based on SIOC, which allows structuring CWE data in a semantically interpretable format. The proprietary CWE data is annotated with the SIOC ontology, thus becoming interoperable with the data from other logical CWEs. Each of the legacy CWEs are required to export their data as SIOC-annotated RDF data through the use of SIOC Exporters. These exporters are software that automatically annotates CWE data with concepts from the SIOC ontology. The SIOC annotated data is accessible through a set of BCS, including the BCS:Search.

Figure 9 shows the services that are used by the SIOC Xplore Widget. The two BCS that are utilised are BCS: Authenticate and BCS:Search.

BCS:Authenticate

BCS:Authenticate is used here to authenticate the user with a CWE, so that they may access data from that CWE. The signature is as follows:

boolean Authenticate (technology, operationType, set of pair-value)

The following outlines how the BCS:Authenticate service is invoked. It authenticates the user 'mary', with the password 'foo', with the CWE located at 'http://www.cwe-projects.eu/bscw/bscw.cgi/'. The specified technology is 'UserManagement' and the operationType is 'AuthenticateUser'.

```
Authenticate (
UserManagement, AuthenticateUser,
      ([username:   mary],   [password:   foo],   [cwe_uri:
      http://www.cwe-projects.eu/bscw/bscw.cgi/]))
```

BCS:Search

The BCS:Search service is used here to query a CWE for data with a particular search criteria. The signature is as follows:

String Search (technology, operationType, set of pair-value)

The following outlines how the BCS:Search service is invoked. It submits the query "SELECT * WHERE.." to the CWE located at 'http://www.cwe-projects.eu/bscw/bscw.cgi/'. The specified CWE technology (see Sect. 5.4) is 'WorkspaceSynchronisation' and the operationType is 'SearchCWE'.

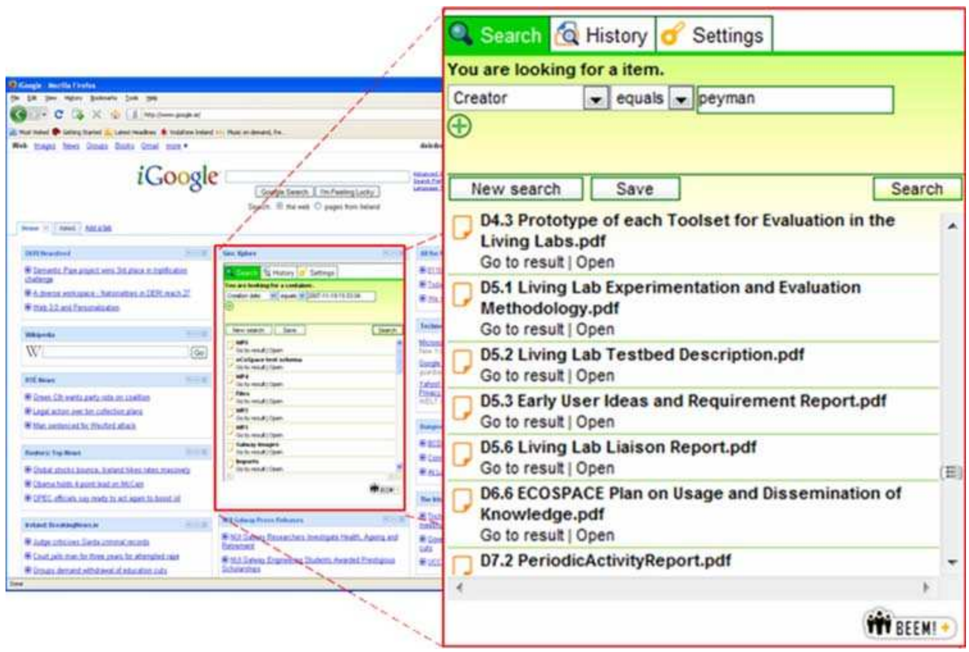

Fig. 8 SIOC Xplore widget, embedded in iGoogle

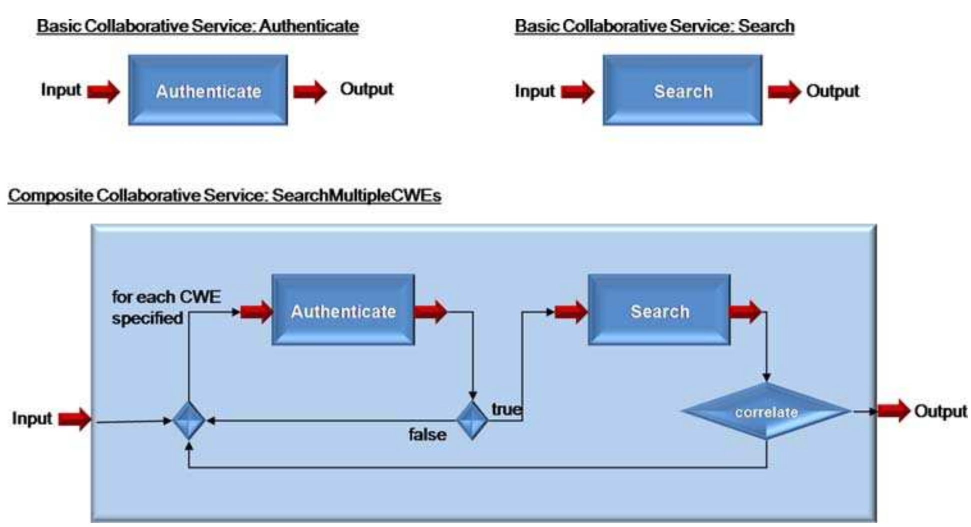

Fig. 9 Basic collaborative services used by the SIOC Xplore widget

```
Search (
      WorkspaceSynchronisation, SearchCWE,
      ([query: SELECT * WHERE..], [cwe_uri: http://www.cwe-
      projects.eu/bscw/bscw.cgi/]))
```

CoCoS:SearchMultipleCWEs

The CoCoS:SearchMultipleCWEs consists of both BCS: Authenticate and BCS:Search, as well as the workflow logic of the service execution, i.e. information related to the sequence/orchestration of the BCSs. As can be seen in Fig. 9 for each CWE specified in the input to CoCoS:SearchMultipleCWEs, the user is first authenticated. If authentication is granted, the specified query is submitted to the CWE. The results of this query are correlated with previous results. When all CWEs specified have been iterated over, the correlated results are returned as output. The signature of this composite collaborative service is as follows:

String SearchMultipleCWEs (set of pair-value)

The following outlines how the CoCoS:SearchMultipleCWEs service is invoked. It submits a list of CWEs to be queried over, complete with authentication details (username and password) and the query to be submitted 'SELECT * WHERE..'.

```
SearchMultipleCWEs (
      [cwe_list: (
            [username: mary], [password: foo], [cwe_uri:
            http://www.cwe-projects.eu/bscw/bscw.cgi/]), (
            [username: maryB], [password: foo], [cwe_uri:
            http://ecospace.withbc.com/bc/bc.cgi/]) ]
      [query: SELECT * WHERE..])
```

Figure 10 demonstrates how the SIOC Xplore Widget's design maps to CERA. A user of the SIOC Xplore Widget finds the widget either at the website supporting the CWE or in a widget gallery and places it on his/her widget panel, desktop, social networking profile, weblog, or any other appropriate location. The first step in using the widget, involves the user setting up their identity. The default authentication process is based on OpenID, thus providing a single identity to engage with each CWE platform. However, SIOC Xplore recognises the need to deal with CWE platforms that are non-compliant with OpenID and this means the user is required to provide CWE-specific authentication details, if they wish to query data from such CWEs.

Once authenticated, the user selects criteria from a dynamic and personalised search menu. The user adds and removes search criteria on demand, and templates are available to facilitate the process. In addition, the SIOC Xplore maintains the user's search history to facilitate efficient querying. This functionality is handled by the SIOC Xplore Broker tool. A specified search is submitted to the CoCoS:SearchMultipleCWEs to be evaluated. The collaborative service, in turn, invokes BCS:Authenticate and BCS:Search, which are defined in terms of the CWA technologies: WorkspaceSynchronisation and UserManagement. The query results are returned to the SIOC Xplore Widget User Interface, where the resultant items' metadata is summarised and presented to the user according to user preferences. A link to the actual item in the original CWE is also provided, so that the user can access the full content from its source, if so desired.

6.2 Workspace synchronisation tool

The Workspace Synchronisation Tool enables CWE users to access remote workspace objects and workspace organisation (i.e. the folder structure), as if they were local CWE objects. The Workspace Synchronisation Tool is integrated into the CWE through the provision of additional interface options in the workspace user interface. These components provide access to the workspaces of remote systems in a transparent way and in the same look and feel as if these were local workspaces. The following scenario, portrayed in Fig. 11, outlines the usage of the Workspace Synchronisation Tool:

> Two companies (A and B) collaborate on a project. They both wish to use their own organisational CWE platforms for the local and shared management of the project. Company A uses BSCW and company B uses BC. Both companies initially create the project workspaces in their local system. Company A then invites

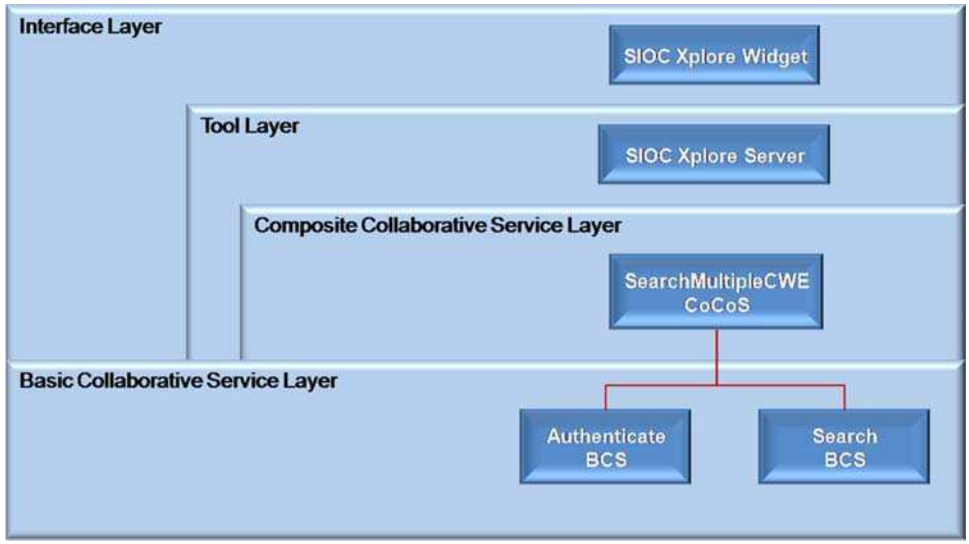

Fig. 10 SIOC Xplore widget's use of CERA

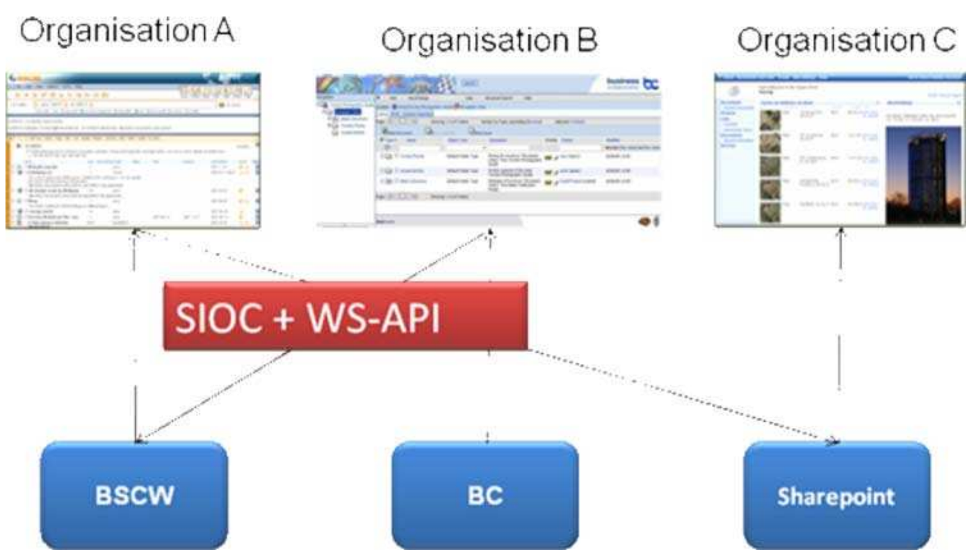

Fig. 11 Using BCS and SIOC for workspace synchronisation

company B to be a user in company A's local workspace, so that the company B has full access to company A's project workspace. Now, using the Workspace Synchronisation Tool, company B may import remote workspace objects from company A's workspace.

In BSCW this is implemented as follows: A user in BSCW creates a new folder, called a shadow folder, by providing the access path to the remote folder as well as the authentication information to access the remote system. This shadow folder is created as a sub folder of the local project workspace. When a user opens this folder a special background colour indicates that the information provided in this folder is not stored in the local system but accessed from a remote server. Otherwise the folder offers the same look and feel the user is used to. Thus he can use the information that is actually stored in a remote system in a transparent way without the need to learn a new system. The functionality offered by this shadow folder is related to the necessary set of services required to access and modify objects. Thus the user cannot use all the advanced features that he may use in his/her local system, e.g. rating or the annotation of objects. Future work will address this issue by the development of a more advanced protocol that enables also the exchange of service capabilities between the different shared workspace systems.

The adopted CERA Content Model, namely SIOC in the Ecospace case, facilitates data interoperability between CWEs. Each of the legacy CWEs are required to export their data as SIOC RDF data through the use of SIOC Exporters. The SIOC annotated data is accessible through a set of BCS, including the BCS: Add, BCS: Delete, BCS: Edit, BCS: Replace, and BCS: Get.

Figure 12 shows the services that are used by the Workspace Synchronisation Tool. As an example, we discuss the

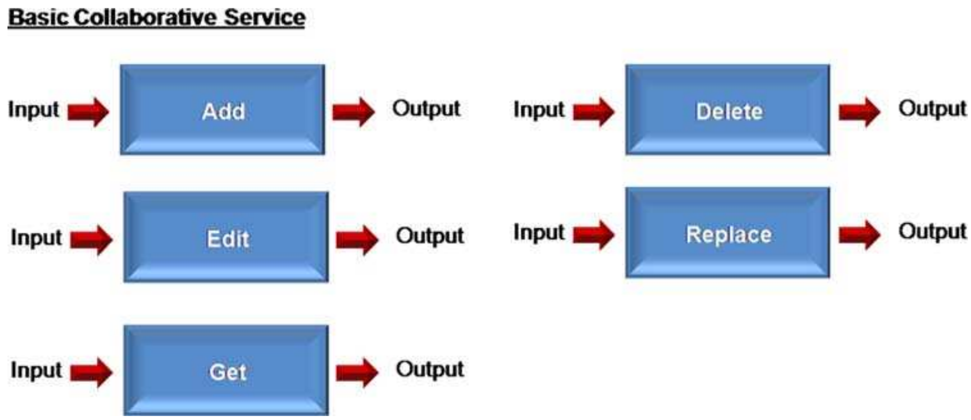

Fig. 12 Basic collaborative services used by the workspace synchronisation tool

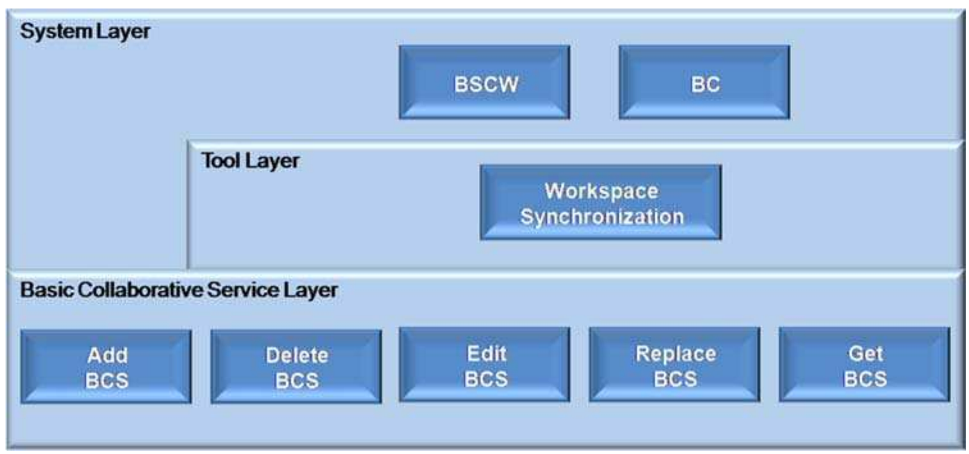

Fig. 13 Workspace synchronisation tool's use of CERA

BCS:Get in more detail. The get basic service is used here to retrieve a remote item or a remote folder structure. The signature is:

Get (technology, operationType, set of pair-value)

The following outlines how the BCS:Get service may be invoked in two different ways by the Workspace Synchronisation Tool.

Get (WorkspaceSynchronisation, GetItem, ([***documentID:*** 4022]))
Get (WorkspaceSynchronisation, GetFolderContent, ([***folderID:*** 2631], [***depth:*** 3]))

The first service call returns an item with a document id '4022'. The second call returns the folder structure of a folder with an id '2631' and a depth of '3'. The folder structure is defined using the Content model; in this case SIOC.

Figure 13 demonstrates how the Workspace Synchronisation Tool maps to CERA. At the system layer, the tool is integrated directly into the CWE system interface. BSCW's implementation of the tool is presented as a Shadow Browser; in BC it is presented as a Semantic Folder. The BCS used by the tool are defined in terms of the CWA technology, namely 'WorkspaceSynchronisation', and are defined in a WSDL file.[7]

7 Evaluation

The Ecospace project has provided an ideal opportunity to evaluate the approach outlined by the initial vision, by bringing together multiple large CWE systems and tools commonly used in the market for e-collaboration. After two years, the reference architecture has matured, having gone through two complete development cycles, which included user testing of the resulting platforms by a large community of users in the project living labs. A third final cycle is foreseen just before the termination of the project.

The user testing at each phase provided valuable input data into the subsequent development cycle, but mainly from a usability perspective and on identifying user needs for additional functionality to be supported. Therefore, it was

[7] http://www.cwe-projects.eu/bscw_resources/wsdl/WorkspaceSynchronisation.wsdl.

Table 5 CERA evaluation methodology

Phase	Step	Designation	Description
Preparation	1	Explanation of evaluation process	The participant is given an overview of the evaluation process, describing what it entails and its purpose
	2	Provision of architecture documentation	The participant is provided with documentation and/or a presentation outlining CERA: its design and some illustrative examples of its use
Analysis	3	Identification of participant's profile	The participant completes a questionnaire that helps identify their participant profile. This will also contribute to the analysis of the responses, thus assisting in the evaluation outcome
	4	Motivation of architecture	The motivation driving the development of CERA is outlined to the participant. This presentation involves the analysis of key requirements in CWEs, and the assessment of common architectural strategies/styles in addressing those needs. Also included is a cost/benefit analysis of the various approaches presented, possibly using examples as support. This enables the identification of implementation dilemmas that may lead to potential abstractions
	5	Scenario identification	The main scenarios that should be supported by CERA are identified. These scenarios are described in sufficient detail to capture the quality attribute whilst eliminating the ambiguity. An initial list of scenarios from the development process of CERA will form the baseline set
Brainstorm	6	Scenario revision	The scenarios identified in step 5 are revised, taking into account the context of the participants participating in the evaluation process. This enables the revision of existing scenarios and the addition of new ones. The scenarios are then prioritised
	7	Architecture discussion	Whether or not the prioritised set of scenarios is supported by CERA is discussed. The result may identify shortcomings, which may require architectural changes. The step also involves the discussion of the impact of those changes and assesses the cost/benefit of such changes
Synthesis	8	Assessment	The results of the architecture discussion are consolidated, taking into account the list of prioritised scenarios. How well CERA fulfilled its quality attributes is examined. The supporting assessment tools may consist of a questionnaire and/or semi-structured interview
	9	Final assessment	It is expected that the evaluation methodology will be executed involving as many participants representatives of the CWE community. This step involves the analysis and synthesis of all the collated data from the various instantiations of the first 8 steps

necessary to define an evaluation methodology that assessed the quality attributes of the CERA reference architecture and its potential risks. The devised methodology was derived from the evaluation methodology [17] used to assess distributed virtual environment systems. The resulting methodology consisted of the same 10 steps, but adapted to the domain of CWE as demonstrated in the outline of Table 5.

The evaluation methodology was executed individually and in a workshop setting, involving a total twenty-five CSCW domain-experts from different backgrounds, including CWE, Semantic Desktop, SOA, and Groupware.

The major findings were as follows:

- The majority of participants had the opinion that architecture plays a vital role in system development and is essential for the successful outcome of a project.
- Most of the non-CWE-expert participants and all of the CWE experts agreed that the BCS Layer as introduced in CERA corresponds to the foundational layer of any CWE system. Moreover all of the participants identified the BCS Layer and the CoCoS Layer as the key building blocks for CERA. Moreover, the majority of participants

agreed that the Semantic Infrastructure Layer can augment and enhance the other layers in CERA.
- The vast majority of the participants responded that the components should be reused from past and existing solutions rather than recreating them. This finding supports the argument that most people are in favour of reusability in CWE development. Based on their CWE background, all of the participants agreed that there is a lot of overlap in existing CWE system in terms of supported functionality, therefore it is feasible and desirable to reuse existing functionality while building new CWE systems.
- 85% of the participants responded that they would very much consider adopting the CERA in CWE related projects. This is one of the key outcomes of the evaluation exercise, as it endorses the rationale of the reference architecture evaluation to verify its adoption within the CWE community.

The next phase of the evaluation process includes the evaluation of CERA within the CWE community and outside the consortium..

Although the results are extremely encouraging, it is foreseen the execution of the methodology with a wider number of experts from the CWE and relevant communities (e.g.: groupware), thus collating more data for analysis.

8 Conclusion and future work

In this paper, we introduced the CERA with the aim of supporting CWA interoperability. The vision of CERA is to support users engaged in common collaborative spaces with similar work processes but different set of CWE tools and systems to seamlessly work and collaborate. We also used the experiences gained from Ecospace to demonstrate how the CERA layers have been instantiated within the project.

In addition to the large community evaluation, and following the W3C standardisation initiative of the SIOC ontology, we plan to explore the standardisation path for other CERA elements, such as the BCS and the composition mechanisms of CoCoS. Interestingly, there have recently been similar efforts and a growing interest in the CWE community towards standardisation (e.g. Core Collaboration Services [25], Oracle Beehive [26])

The Ecospace project continues to add and explore CWE functionalities, building upon and extending the existing work on the reference architecture. One direction we foresee to this end is related to the development of a CWE functional language to support automated creation of tools and services based on the CoCoS, thus exploring its full potential and leading towards user tailorability of CWE functionalities at run-time.

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